Qadam / ArcSolo

Test scenarios, limitations and privacy — Build with AI, Track 1

Prepared 19 October 2026. Project documentation: English. Pitch: Russian. This report accompanies the working prototype, automated OSM collector, source repository and presentation.

Verification summary

103 engine/API/Worker tests; 6 graph-collector tests; 160 combinations of location, time, interests and weather; 100 combinations checking route/card agreement; 25/25 acceptance scenarios; 98/98 browser checks (73 planner + 25 companion). These are deterministic engineering checks, not a user study or a live-model quality benchmark.

The targeted intent audit improves from 3/10 to 10/10; 12 additional authored paraphrases pass. These are not representative accuracy scores. Version 5 extends paths to five collected areas, handles word durations and Kazakh deadlines, pauses ambiguous timing, corrects incoming-leg times on cards, and expands local evidence. Three source-backed stories, recovery and return buffers remain. See docs/NEXT_STEPS.md for evidence and limits.

Adaptive walk companion

Version 6 recalculates unfinished stops from a manually confirmed point while preserving the original return destination, absolute deadline and reserve. The 17 companion code tests cover elapsed time, unavailability, weather, hours, return-only and unreachable states. An additional 25 browser checks cover actual/rehearsal modes, three languages, mobile/keyboard use and privacy. A rehearsal demonstrates an unavailable café, 15 minutes lost and rain without an outdoor field test. Actual mode uses the device clock on each action; it has no background navigation or GPS tracking. These are engineering checks, not measured user benefit.

Data provenance

300 named POIs across five Astana neighborhoods, automatically collected from OpenStreetMap on 18 October 2026. 102 have opening hours; 26 have wheelchair tags. Descriptions are explicitly labelled summaries of source category, cuisine and street tags. A separate enrichment layer adds 32 first-party facts across 18 places (19 enriched records including access-only evidence), three supported weekly-hours overrides and two campus-access exclusions. Source IDs, original tags, collection date and metadata checksum are preserved. OSM-derived data is licensed ODbL 1.0, © OpenStreetMap contributors.

RAG approach

Automated source collection BM25 and multilingual concept retrieval hard geographic/time/context checks recommendations and a short route optional Gemini structured generation validation of POI and citation IDs, output bounds and common unsupported claims. Gemini writes personalized explanations, activities and discovery missions from retrieved facts and normalized context. Six live scenarios are retained in docs/live-rag-evaluation.json; deployment verification is in docs/live-generation-check.json. Citation checks do not prove semantic support for every generated sentence. These smoke checks establish connectivity and validated output, not general model quality. Mock tests cover malformed and unsupported responses; provider failures fall back to explicitly labelled

templates.

Acceptance scenarios

story-expo-space-and-symbols — PASS

Expected: Both source-backed chapters fit; every factual chapter resolves to its exact evidence record.

Observed: 2 recommendations; route estimate 88 / 120 minutes. Places: Mega Silk Way, Тәуелсіздіктің 25 жылдығы.

story-symbols-in-sculpture — PASS

Expected: Both source-backed chapters fit; every factual chapter resolves to its exact evidence record.

Observed: 2 recommendations; route estimate 96 / 120 minutes. Places: Бәйтерек, Ғашықтар саябағы.

story-ancient-life-modern-city — PASS

Expected: Both source-backed chapters fit; every factual chapter resolves to its exact evidence record.

Observed: 2 recommendations; route estimate 104 / 120 minutes. Places: Қазақстан Республикасының Ұлттық Мұражайы, Президент саябағы.

mapped-expo — PASS

Expected: Route uses mapped EXPO geometry while retaining explicit estimated endpoint connections.

Observed: 2 recommendations; route estimate 53 / 60 minutes. Places: Лепим и варим, Бейбітшілік пен Келісім аллеясы.

compound-duration — PASS

Expected: Effective time is 90 minutes and walking cap is 10 minutes.

Observed: 3 recommendations; route estimate 82 / 90 minutes. Places: Mega Silk Way, Luckee Yu, Gasteria.

deadline-buffer — PASS

Expected: The deadline forces a return; visits, walking and ten-minute reserve fit before 15:00.

Observed: 2 recommendations; route estimate 48 / 50 minutes. Places: Лепим и варим, Luckee Yu.

expired-deadline — PASS

Expected: Past deadline yields no stops and a visible deadline warning.

Observed: 0 recommendations; route estimate 0 / 0 minutes. No places returned.

recovery-museum — PASS

Expected: Empty short museum plan offers a recomputed time increase without relaxing the walking limit.

Observed: 0 recommendations; route estimate 0 / 15 minutes. No places returned.

no-coffee-en — PASS

Expected: Park request overrides stale coffee interest; cafes excluded.

Observed: 1 recommendations; route estimate 52 / 90 minutes. Places: Бейбітшілік пен Келісім аллеясы.

no-coffee-ru — PASS

Expected: Park request overrides stale coffee interest; cafes excluded.

Observed: 1 recommendations; route estimate 52 / 90 minutes. Places: Бейбітшілік пен Келісім аллеясы.

no-coffee-kk — PASS

Expected: Park request overrides stale coffee interest; cafes excluded.

Observed: 1 recommendations; route estimate 52 / 90 minutes. Places: Бейбітшілік пен Келісім аллеясы.

indoor-only — PASS

Expected: No outdoor category in an indoor-only itinerary.

Observed: 3 recommendations; route estimate 90 / 90 minutes. Places: Лепим и варим, Хан Шатыр, Ozyurt.

short-park-visit — PASS

Expected: A nearby park can fit as a labelled short visit within 15 minutes.

Observed: 1 recommendations; route estimate 10 / 15 minutes. Places: Жетісу саябағы.

group-coverage — PASS

Expected: All three chosen interests covered by a feasible shared route.

Observed: 3 recommendations; route estimate 93 / 120 minutes. Places: Лепим и варим, Тәуелсіздіктің 25 жылдығы, Бейбітшілік пен Келісім аллеясы.

monday-museum — PASS

Expected: National Museum excluded on Monday using official hours.

Observed: 0 recommendations; route estimate 0 / 120 minutes. No places returned.

restricted-campus — PASS

Expected: NU Park and Japanese Garden excluded due to campus access restrictions.

Observed: 1 recommendations; route estimate 52 / 120 minutes. Places: Бейбітшілік пен Келісім аллеясы.

baiterek-evidence — PASS

Expected: Baiterek retrieved with official architectural evidence and an exterior visit scope.

Observed: 1 recommendations; route estimate 22 / 60 minutes. Places: Бәйтерек.

expo-hour — PASS

Expected: 1–3 nearby options; total route including return 60 minutes.

Observed: 2 recommendations; route estimate 53 / 60 minutes. Places: Лепим и варим, Бейбітшілік пен Келісім аллеясы.

quiet-coffee — PASS

Expected: Cafés only; quietness and sockets explicitly unverified.

Observed: 1 recommendations; route estimate 30 / 60 minutes. Places: Degirmen.

family-museum — PASS

Expected: Family candidate categories; no invented interactive exhibits.

Observed: 2 recommendations; route estimate 104 / 120 minutes. Places: Қазақстан Республикасының Ұлттық Мұражайы, Президент саябағы.

rainy-khan — PASS

Expected: Indoor categories; weather labelled user-selected.

Observed: 3 recommendations; route estimate 90 / 90 minutes. Places: Лепим и варим, Хан Шатыр, Ozyurt.

too-little-time — PASS

Expected: No invented match when nothing fits.

Observed: 0 recommendations; route estimate 0 / 10 minutes. No places returned.

wheelchair — PASS

Expected: Only explicit wheelchair=yes; route access still unverified.

Observed: 2 recommendations; route estimate 56 / 60 minutes. Places: Керуен, Избушка.

airport-night — PASS

Expected: Airport buffer and night safety warnings; no flight guarantee.

Observed: 3 recommendations; route estimate 113 / 120 minutes. Places: Coco, завтрак бар, Тәуелсіздіктің 25 жылдығы.

outside-coverage — PASS

Expected: Empty result outside collected Astana areas.

Observed: 0 recommendations; route estimate 0 / 60 minutes. No places returned.

Scenarios simulate 19 October 2026, 14:00 Astana (UTC+5), except the night case and the family museum case on Tuesday 20 October. Results are snapshot-based, not claims of verified current availability.

Limits and ethics

Generated prose is not automatically proven true. Place/citation membership, output bounds, exterior scopes and common unsupported claims are checked, but semantic errors can remain. Source facts are displayed separately.

Six live generation scenarios in EN/RU/KK are retained for human review. This is not a statistically meaningful model-quality benchmark or user study. Mock responses are used for repeatable browser checks.

OSM coverage is limited to five neighborhoods. Primary-source enrichment is manually researched and separately attributed; it is not a live source crawl on every request.

Published schedules can be overridden by holidays or exceptional closures. Source-check dates are not on-site verification dates. Complex or seasonal hours remain unknown unless explicitly supported for the relevant scope.

Five collected areas use connected OSM pedestrian geometry. Unknown endpoint connections and areas outside that graph remain estimated as straight-line distance × 1.3. Speeds are 75 m/min, or 55 in gentle mode. Missing connections between snapped components are rejected. Mapped entrances and exterior viewpoints are not on-site verified; sightlines, crossings, queues, closures and step-free access are not guaranteed.

Return deadlines reserve time inside the budget; they do not guarantee punctual arrival. Recovery checks 15-minute increases up to 120 extra minutes and selected later departures, not every possible alternative. Story pairings are editorial and do not imply a shared historical origin.

Short route pauses are planning choices, not promises about food preparation or admission. Full-meal and indoor-attraction visits may need additional time.

Group coverage means matching selected categories, not equal enjoyment or a validated preference model.

The companion updates actual time on user actions, not continuously or in the background. The selected POI is an approximate confirmed point, not a GPS fix. Rehearsal simulates planned durations. It supports round trips only, and sessions disappear on reload. Unavailability reports affect only this session; they do not update shared venue facts.

Weather is manual. Ratings, live events, prices, sockets and quietness have no live integration.

Natural-language parsing recognizes a limited multilingual vocabulary, common negation patterns and durations. It is heuristic, not general language understanding. Use the structured controls for reliable constraints.

Native-speaker review of Kazakh and real participant feedback are still needed.

Precise geolocation is requested only after a user click and browser permission. It lives in page memory; a clear-location action removes it. Group interests, route edits, up to ten undo snapshots, mission checkmarks and a bounded generation cache are also page-memory-only. Origin-dependent graph snaps, shortest-path trees and route geometry are bounded page-memory caches; clearing a custom location clears the router caches. Deadlines, reserve choices, recovery history and selected story remain in page memory. The app writes no location history, cookies, account database, localStorage, sessionStorage or analytics events. Preset landmarks are available without location permission.

Recommendation ranking and route planning run locally in the browser. Map geometry is included with the app; the browser makes no external map-tile requests. The optional generation request sends only selected public POI IDs, estimated visit lengths, language, mood, category interests, up to five anonymous group preferences weather, indoor/outdoor intent, excluded interest categories, bounded time budget and three booleans (family, quiet, sockets). The server reconstructs trusted records and sends those public records to Gemini. It never forwards the user's precise origin or raw query. Hosting infrastructure and the model provider may still process network metadata. Opening a source link is an explicit external navigation.

POI records are untrusted input. Text is escaped before rendering; map source links are restricted to canonical OSM record URLs. The model is told not to follow source instructions, and its output must contain one bounded explanation, activity and mission per selected POI, with valid citation IDs belonging to that place. Exterior-only stops cannot cite interior exhibits. Fabricated IDs, unsupported citations, duplicate records, malformed JSON, common unsupported operational claims, HTTP errors and timeouts trigger the labelled fallback. Citation validation does not prove semantic support for every generated claim; generated prose remains visibly labelled and source facts are shown separately. Venue names, coordinates, hours and accessibility come from source records, never the model.

Unknown opening hours are not “open now.” The parser handles a limited weekly subset and treats unsupported expressions as unknown. Availability is not guaranteed even when a visit fits the published schedule. Accessibility tags are not verified step-free routes. Sockets, quietness, night security and lighting remain unknown without evidence. Official descriptions support some museum interactive exhibits, but current availability is still unverified. Activity suggestions are separately labelled.

Local serving is loopback-only. API responses are no-store; bodies are limited to 8 KiB; cross-origin browser writes are rejected. The public judge Site has paid generation disabled. A future public live-model service would need authentication/rate limits and budget controls. Keep `.env` ignored and configure hosted secrets through the hosting provider, never client JavaScript.

Future monetization must not bypass relevance or feasibility filters. Sponsored results must be clearly labelled. Location histories are not part of the business model.

The public judge preview sets `QADAM_PUBLIC_DEMO=true`, returns a labelled retrieval/template response before any provider request, and has no hosted Gemini key. Live generation remains opt-in in a local installation with its own server-side key. Neither public-demo status nor provider keys can be changed through request bodies.

Adaptive companion

Confirmed stop positions, finished/unavailable place IDs, the original return point, fixed deadline, manual weather reports and up to 20 rehearsal undo states remain in page memory. No location watcher or background timer is started. Ending the companion clears its session and restores the original plan; reloading clears walk progress. Actual time is refreshed on user actions and rounded up to the next minute. Rehearsal advances explicitly simulated travel/visit/delay time and can undo it; actual mode cannot rewind time.

The generation boundary is unchanged: only the selected remaining public places and existing normalized context are sent. Session history, event types, current/return coordinates, exact clock and deadline are not sent. Selected place IDs can still reveal a coarse area of interest. An unavailable-place report excludes that place only from this walk and is not published as a source fact. Routes, deadlines and reserve are estimates, not arrival guarantees. If no further visit fits, only the return is shown; missing connectivity, lateness or a walking-cap conflict is disclosed.

Submission status

Repository: https://github.com/Csadow/qadam-astana (public; anonymous access verified). Hosted demo: https://qadam-arcsolo-astana.candy-ball-8297.chatgpt.site (public judge preview; paid generation disabled). A source archive is prepared as an alternative. The captioned MP4 is ready, but it has not been uploaded to YouTube. YouTube upload and final form submission remain participant actions. Participant testing is deferred.